Mayu Tatsumi

+1 (929) 469-0988 | mtatsumi@g.hmc.edu | linkedin.com/in/mayutatsumi

Education

Harvey Mudd College — Claremont, CA

B.S. Engineering — *Expected May 2027*

GPA 3.75, Dean's List

Coursework: Systems & Signals, Control Theory, RF Circuit Design, Microprocessor Design, Digital Electronics & Computer Engineering (Digital Signal Processing, Analog Circuit Design)

Skills

Technical Languages: SystemVerilog, Python, C, Linux
Programs: MATLAB/Simulink, LabVIEW, LTSpice, KiCAD, Lattice Radiant (iCE40), Segger Embedded Studio (STM32), Intel Quartus
Tools: VNA, Oscilloscope, Logic Analyzer, Soldering, Multimeter
Languages: English, Japanese, Spanish (Basic), Korean (Basic)

Relevant Experience

Yaskawa America, Inc. Solectria Solar — Lawrence, MA

Control Systems Engineering Intern — *May 2026 – Aug 2026*

- Modeled, extracted, and compared frequency response of MATLAB Simulink model and physical lab PV inverter utilizing small-signal perturbation and control systems.
- Audited and began migrating a 60+ package Jenkins/Debian build pipeline from 32-bit to 64-bit NXP i.MX ARM chip; converted C binaries to portable shell scripts and validated changes against live hardware.

TechMagic, Inc. — Tokyo, JPN

Controls Systems Intern — *May 2025 – June 2025*

- Designed, implemented, and debugged C firmware for an NXP i.MX8M series processor, enabling a nationwide firmware update for deployed robots.
- Designed tests and edge cases to verify firmware reliability outlined in design reviews.
- Side Projects: Remaking GUI in Flutter, creating MQTT modules, and reconfiguring the Watchdog system.

Departments of Engineering and CS, Harvey Mudd College — Claremont, CA

Teaching Assistant — *Aug 2024 – Present*

- Tutored students, one-on-one, for courses Signals & Systems and Control Theory.
- Graded coursework for Applied Math for Engineering, Experimental Engineering, Engineering Systems, and CS42: Principles and Practice.

Projects

Fluorescence-Guided Surgery Camera Module — PATH Imaging, Inc. Clinic, Harvey Mudd College — Aug 2026 – Present

- Developed custom firmware in Linux to interface with IR/FL and RGB cameras utilizing libraries such as IOCTL and V4L2.
- Modeled sensors and lenses using homography to optimize image alignment, quantified alignment using Mutual Information metrics.
- Designed test bays and testing procedures to standardize and verify imaging and processing accuracy.

Karaoke Machine — E155, Harvey Mudd College — Nov-Dec 2025

- Defined technical specs and requirements for a karaoke machine that plays a song, displays lyrics, detects the singer's notes, and computes a grade at the end,
- Designed filter stages and coefficients for CIC, FIR, and half-band filters, and implemented them on a Lattice iCE40UP FPGA board using SystemVerilog RTL and Lattice Radiant software.
- Developed custom DMA and SPI peripheral drivers on an STM32-Nucleo 32 MCU board, utilizing interrupts in C and SEGGER.
- Interfaced with LCD, speaker, and PDM MEMS digital microphone hardware components.

Transdermal Device Data Acquisition System — Indomo, Inc. Clinic, Harvey Mudd College — Jan 2026 – May 2026

- Designed signal filters and wrote Arduino code to interface with 9-DoF IMU and provide real-time calculated shake intensity and rotation over time in MATLAB.

Multicycle RISC-V Processor — E85, Harvey Mudd College — Nov – Dec 2024

- Designed and implemented a multicycle RISC-V processor in SystemVerilog, adapting single-cycle datapath components and developing custom control logic for a unified memory architecture.
- Simulated, debugged, and verified processor functionality using Intel Quartus, achieving full test coverage.

Professional Development & Leadership

Asian-Pacific Islander Sponsorship Program Sponsor — Harvey Mudd College — Aug 2023 – Present

MuddLife Co-Director — Harvey Mudd Executive Board — Aug 2025 – May 2026

Next-Gen Innovators Mentorship Mentee — Apple Inc. — Aug 2025 – May 2026